Ch. 7.3 HW

Principles of Microeconomics

October 8, 2025

Problems

Quick Quiz #7-10, Problems & Applications #6-8

Quick Quiz

7. Isabelle values her time at $60 an hour. She spends 2 hours giving Jayla a massage. Jayla was willing to pay as much as $300 for the massage, but they negotiated a price of $200. In this transaction,

   (a) consumer surplus is $20 larger than producer surplus.

   (b) consumer surplus is $40 larger than producer surplus.

   (c) producer surplus is $20 larger than consumer surplus.

   (d) producer surplus is $40 larger than consumer surplus.

8. An efficient allocation of resources maximizes

   (a) consumer surplus.

   (b) producer surplus.

   (c) consumer surplus plus producer surplus.

   (d) consumer surplus minus producer surplus.

9. When a market is in equilibrium, the buyers are those with the _____ willingness to pay, and the sellers are those with the _____ costs.

   (a) highest; highest

   (b) highest; lowest

   (c) lowest; highest

   (d) lowest; lowest

10. Producing a quantity larger than the equilibrium of supply and demand is inefficient because the marginal buyer's willingness to pay is

    (a) negative.
    (b) zero.
    (c) positive but less than the marginal seller's cost.
    (d) positive and greater than the marginal seller's cost.

Problems and Applications

6. Consider a market in which Bert from problem 4 is the buyer and Ernie from problem 5 is the seller.

    (a) Use Ernie's supply schedule and Bert's demand schedule to find the quantity supplied and quantity demanded at prices of $2, $4, and $6. Which of these prices brings supply and demand into equilibrium?
    (b) What are consumer surplus, producer surplus, and total surplus in this equilibrium?
    (c) If Ernie produced and Bert consumed one fewer bottle of water, what would happen to total surplus?
    (d) If Ernie produced and Bert consumed one additional bottle of water, what would happen to total surplus?

7. The cost of producing flat-screen TVs has fallen over the past decade. Let's consider some implications of this change.

    (a) Draw a supply-and-demand diagram to show the effect of falling production costs on the price and quantity of flat-screen TVs sold.
    (b) In your diagram, show what happens to consumer surplus and producer surplus.
    (c) Suppose the supply of flat screen TVs is very elastic. Who benefits most from falling production costs – consumers or producers of these TVs?

8. Four consumers are willing to pay these amounts for haircuts:

Gloria: $35	Jay: $10	Claire: $40	Phil: $25

    Four haircutting businesses have these costs:

Firm A: $15	Firm B: $30	Firm C: $20	Firm D: $10

    Each firm can give, at most, one haircut. To achieve efficiency, how many haircuts should be given? Which businesses should cut hair and which consumers should have their hair cut? How large is the maximum possible total surplus?